

Architectural Particular Specification
TURNTABLE
SECTION 11.1

Rev	Date	Description
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SECTION 11.1

TURNTABLE

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The work shall include the design, supply, delivery, installation, testing and commissioning for operation of vehicle turntable with all necessary equipment including control panel at the Carpark / Loading/Unloading Area of the proposed data centre development as shown in the drawings.

1.02 REFERENCE

- A. The turntable shall comply Section C11 'Vehicle Turntable' at General Specification for Mechanical Installation issued by ArchSD.

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

- A. The turntable shall be supported by a centre pivot on roller bearing and castors at the circumference. Number of castors at the circumference varies according to the location condition.
- B. These special designed castors and centre pivot shall enable smooth rotation of turntable. Top deck is formed of steel chequered plates bolted to the framework. An edge ring of steel angle is fixed to the concrete around the turntable to ensure a uniform gap between the turntable and the floor.
- C. Framework of the turntable shall be constructed with steel sections and profiles welded/ bolted together. The framework shall be made of hot-dip galvanised steel.
- D. The main contractor shall also provide all other necessary components including but not limited to the followings:
 - 1. Concrete pit accommodating the turntable and drive unit, and with kerb angle ring anchoring rods fitted;
 - 2. Drainage of concrete pit;
 - 3. Concealed electrical conduit from the drive unit compartment to the wall-mounted control panel; and
 - 4. T. P. & N. Isolator next to the control panel
- E. Finishes
 - 1. All steelwork shall be painted with one high quality primer coat and two top coats for surface protection.
 - 2. Cover plate shall be painted to in suitable paint system in matching colour with carpark floor finish.

F. Operation Requirements

1. Control

Wall-mounted control panel with on/off key switch, deadman-type control buttons, emergency stop button. Frequency controller for continuously variable speed regulation.

2. Drive

The turntable is driven by an electro-mechanical drive unit composed of a motorgearbox- friction wheel assembly and situated at one side of turntable.

3. Rotating Speed

Approximately 0.75 r.p.m.

4. Rotataion

360° clockwise or anti-clockwise

5. Load bearing capacity

Minimum rated load of turntable shall be 30 tons.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

- A. Detailed shop drawings showing equipment layout, dimension and locations of all major equipment and electrical power requirement etc. shall be submitted for approval.
- B. Manufacturer's catalogues, job reference, test certificate, installation method and material samples for all relevant materials and equipments shall be submitted to the Architect for approval. The extent of information or samples required shall be adequate and sufficient to demonstrate that the proposed system and materials are in compliance with the contract drawings and specification.
- C. No materials and systems ordering shall proceed prior to the Architect' confirmation of no adverse comment on the submitted information or samples. It is the contractor's sole responsibility to ensure that the required materials or systems can be ordered according to the contractor's master programme. No claim for any additional time or cost due to any delayed submission will be entertained.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

Package and store materials in a manner that will prevent damage, distortion, breakage or structural weakening. Replace any materials damage during manufacture, shipping, storage, or installation.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

Upon completion of the work, all equipment shall be tested and commissioned by the manufacturer's own engineer to ensure satisfactory operation. The works shall also be tested and commissioned to the architect's and any relevant government's satisfaction such as EMSD. Similar testing shall be carried out at the end of the maintenance period.

A commissioning certificate with Operation and Maintenance manual shall be submitted.

A staff training session to the building management for the operation and maintenance shall be conducted upon the Architect's notice within the DLP period.

1.12 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

The contractor shall guarantee the turntable for a period 5 years and submit a written warranty in joint name with the manufacturer for the materials, workmanship and performance of the turntable to the Architect for review.

No bulk materials and systems ordering work shall proceed prior to the Architect's confirmation of no adverse comment on the submitted warranty content and format.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER

Subject to compliance with the Project Contract Documents, provide panel ceilings and required accessories from one or more of the following manufacturers, or equivalent approved by the Architect:

A. Arrowtech Engineering (H.K.) Co., Ltd.
B7, 9/F., Merit Industrial Centre,
No. 94 To Kwa Wan Road,
Kowloon, Hong Kong
Tel: (852) 2364 0727
Email: alex.to@arrowtech.com.hk

B. Equivalent approved by the Architect

2.02 MATERIALS / SYSTEMS

A. Vehicle Turntable and Associated Accessories
Performance requirement refer to **1.04**

B. Motor Pit and Sump Pit

C. Electrical Installations

2.03 ANCILLARY MATERIALS

(NOT USED)

2.04 FABRICATION

(NOT USED)

2.05 FINISHES/FINISHING TREATMENTS

(NOT USED)

2.06 STORAGE

(NOT USED)

2.07 WORKMANSHIP

(NOT USED)

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

- A. The driving system and supporting castors shall be visually inspected before erection of cover plates.
- B. The system shall be functional tested with the specified loading capacity after completion.

3.02 PREPARATION

(NOT USED)

3.03 INSTALLATION / APPLICATION

- A. Main Frame Construction
 - 1. The clearance between the turntable pit and the turntable circular edge shall not be greater than 10 mm during operation.
 - 2. All steelwork shall be primed before erection and the under coat and finishing coat shall be coal tar epoxy paint. The priming and painting shall be conducted in accordance with the requirements in Section 4.1 of the Architectural Particular Specification.
 - 3. Castors shall be correctly located and the pit base is reasonable level for supporting the turntable.
- B. Supporting Castors
 - 1. Each castor shall consist of a cast steel wheel running on a stainless steel shaft with self-lubricating bearing, the bearing shall be properly sealed to prevent entry of moisture. The wheels shall be supported on stainless steel wheel brackets to be mounted onto the pit base.
 - 2. Provision shall be made to ensure all castor wheels shall have good contact and match with the running rail as stated below.
 - 3. Running rail(s) of cold rolled steel shall be installed on the underside of the turntable for each set of supporting castors. The dimensions of each rail track shall conform to the dimensions and location of the respective castor wheels underneath the turntable.
- C. Building service installation shall be provided in accordance with the Drawings and Specification. The installation shall also include the followings:
 - 1. The Contractor shall be responsible for the supply and installation of all building service installation including but not limited to the switchgears, cables, cable connections and all other necessary accessories.

3.04 TOLERANCES (NOT USED)

3.05 CLEANING (NOT USED)

3.06 PROTECTION (NOT USED)

3.07 MAINTENANCE AGREEMENT

A. Free Maintenance

The Contractor shall furnish maintenance free of charge for the entire installation for a period of twelve months following the date of Substantial Completion. The extent of work required to be carried out should cover at least, but not limited to the following:

1. Planned Maintenance

- a. To be responsible for any repairs necessary to maintain the installation in good and safe working order at all times.
- b. To dispatch competent and specially trained workmen once monthly during normal working hours to maintain the turntable where applicable.
- c. To supply all lubricants, cleaning materials, rope preservatives etc. required.
- d. Replace all burnt out lamp bulbs/tubes with bulbs/tubes of correct rating.
- e. To provide, repair or replace at no additional cost to the Employer such mechanical and electrical parts of the installation required for the safe and efficient operation of the installation.

2. Emergency Maintenance

- a. To provide a "call-out" service during and outside normal working hours to carry out emergency maintenance by competent workman.
- b. To attend to any breakdowns reported to him by telephone, (or other means) with the utmost effort and in no circumstance attendance to the breakdown shall exceed 30 minutes after the call is received.

3. Statutory Maintenance

To carry out periodic testing and examination of safety equipment as may be required by the provisions of any enactment in force relating thereto or of any enactment, regulation or by-law of any local or other duly constituted authority which may be applicable to such tests and to provide such copies of the test certificate, duly signed by a Registered Lift Engineer, as may be required.

Provided always that any renewals or repairs necessitated by reason of negligence or misuse of the equipment or by reason of any other cause beyond the Contractor's control with the exception of wear and tear shall be carried out by the Contractor if so required by the Employer at an additional cost to be negotiated by both parties.

All work under this maintenance provision shall be performed by competent personnel under the supervision and in the direct employ of the Contractor.

B. Fee Proposal for Maintenance after Free Maintenance Period

The tenderer shall submit with his tender a fee proposal for undertaking maintenance of the installation provided under the Contract for a period of five (5) years from the date of expiration of the Free Maintenance Period. This maintenance proposal shall be prepared on a fully comprehensive maintenance basis including all necessary materials and skilled labours.

C. Training

1. The Contractor shall provide adequate training to employer's personnel regarding operation and maintenance of the installation.
2. Complete, printed or typed instruction booklets shall be provided covering maintenance, operation and adjustment of each piece of equipment.

3. Training shall cover at least, but not be limited to the following:
- (i) familiarization with the equipment supplied;
 - (ii) technical details with regard to control principles and techniques employed;
 - (iii) installation and alignment technique;
 - (iv) testing and adjusting procedures;
 - (v) routine maintenance and overhaul.

END OF SECTION